

FARMVILLE, VA, USA

WMO: 724017

Lat: 37.358N

Lon: 78.438W

Elev: 417

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	18.8	-4.0	4.5	26.4	1.2	5.9	28.3	19.4	49.4	17.6	48.4	1.7	290	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	95.4	75.5	92.7	75.4	90.4	74.6	80.4	89.4	78.6	87.4	77.4	86.3	5.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	151.0	84.9	75.6	135.9	82.2	74.9	132.8	81.5	44.2	89.3	42.3	88.5	41.1	86.0	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	13.5	11.6	DB	5.0	98.8	7.1	3.9	0.0	101.6	-4.2	103.9	-8.1	106.1	-13.3	109.0	
			WB	4.0	81.6	6.8	2.3	-0.9	83.3	-4.9	84.7	-8.7	86.0	-13.7	87.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	38.3	40.7	48.7	58.4	66.3	74.3	78.5	77.1	70.0	59.0	48.2	41.7
	DBStd	16.29	10.54	9.72	9.87	8.79	7.49	5.90	4.69	4.85	6.50	8.13	9.14	9.13
	HDD50	1290	388	288	148	20	0	0	0	0	0	14	142	290
	HDD65	3824	827	682	513	236	79	6	0	0	28	223	507	723
	CDD50	4402	26	26	106	272	506	730	882	839	601	292	89	33
	CDD65	1461	1	0	7	37	120	286	417	374	179	36	3	1
	CDH74	14668	3	6	92	500	1219	2862	4273	3677	1600	393	35	8
	CDH80	6133	0	1	12	139	388	1202	2029	1665	589	106	2	0
Wind	WSAvg	3.9	4.5	4.5	5.0	5.1	4.0	3.4	3.0	2.8	3.3	3.4	3.8	3.8
Precipitation	PrecAvg	43.2	3.5	3.3	4.1	3.0	4.1	3.4	4.4	3.7	3.7	3.5	3.2	3.3
	PrecMax	57.8	8.1	6.7	9.4	7.2	8.7	9.1	8.0	11.3	10.7	12.2	9.6	8.0
	PrecMin	31.6	0.9	0.6	0.4	0.2	1.4	0.5	1.1	0.4	0.1	0.0	0.4	0.3
	PrecStd	6.9	1.7	1.6	1.9	1.5	1.7	2.2	2.0	2.3	2.6	2.6	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	73.1	81.7	89.5	90.7	97.2	100.1	98.9	93.3	88.2	79.0	73.1
		MCWB	61.3	58.7	61.8	66.8	73.1	75.7	77.0	77.7	72.2	70.8	63.7	61.5
	2%	DB	65.9	67.7	76.3	83.8	87.9	92.7	96.8	94.5	90.3	82.2	73.0	66.4
		MCWB	58.9	55.1	59.7	64.9	71.4	76.1	76.1	76.4	72.3	67.9	61.6	60.8
	5%	DB	61.4	63.0	72.0	80.8	84.2	90.1	92.8	91.2	87.0	78.7	69.8	62.9
		MCWB	53.4	52.1	56.9	62.9	70.0	74.8	75.7	75.4	71.7	66.4	60.2	55.8
	10%	DB	55.4	57.2	66.4	75.1	81.4	87.6	90.3	89.3	82.5	73.4	64.7	57.3
		MCWB	48.1	48.2	54.8	60.9	68.4	73.6	75.3	74.6	70.1	63.6	56.9	50.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	62.7	65.9	71.4	76.2	81.0	82.2	82.6	78.5	75.4	68.9	66.3
		MCDB	67.9	69.3	75.6	81.8	85.9	91.3	93.0	90.8	85.0	79.8	72.3	69.7
	2%	WB	60.3	59.1	62.9	68.4	74.0	78.5	80.5	80.2	76.1	72.0	65.7	62.3
		MCDB	65.0	64.7	71.4	78.4	83.3	87.6	90.0	88.6	82.4	79.0	70.0	65.6
	5%	WB	55.2	53.9	59.9	66.0	72.1	77.0	78.6	78.4	74.3	68.9	62.5	57.2
		MCDB	60.1	60.4	68.7	74.3	81.1	85.9	87.5	87.5	81.1	75.2	67.2	61.8
	10%	WB	49.2	48.9	56.7	63.4	70.2	75.4	77.4	76.9	72.8	65.9	58.6	51.8
		MCDB	54.9	56.4	64.8	71.6	78.3	83.9	86.3	85.6	79.9	72.1	64.3	55.9
Mean Daily Temperature Range	5% DB	MDBR	20.7	22.4	24.1	25.0	22.9	21.7	21.0	21.4	22.0	24.3	24.1	21.1
		MCDBR	25.9	29.5	29.3	30.1	26.1	24.6	24.9	24.6	26.6	28.3	29.0	26.0
		MCWBR	18.7	19.5	16.1	14.2	12.4	10.4	9.4	9.6	11.2	15.2	18.3	18.6
	5% WB	MCDBR	22.7	26.0	24.6	23.9	22.4	21.4	20.8	20.4	20.1	22.3	21.7	21.9
		MCWBR	19.1	20.1	16.0	13.7	11.8	10.3	9.4	9.3	10.2	13.6	16.5	18.3
Clear-Sky Solar Irradiance	taub	0.298	0.313	0.347	0.384	0.430	0.477	0.508	0.506	0.403	0.358	0.319	0.302	
	taud	2.526	2.474	2.421	2.334	2.222	2.139	2.075	2.062	2.341	2.456	2.525	2.552	
	Ebn at Noon	285	293	291	284	272	258	249	246	268	272	274	276	
	Edh at Noon	25	29	34	39	45	49	52	51	36	30	25	23	
All-Sky Solar Radiation	RadAvg	756	994	1316	1635	1832	2003	1892	1706	1405	1119	836	657	
	RadStd	56	117	90	125	156	134	108	85	144	130	60	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	+1.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+265	+167

Nomenclature: See separate page